



Test : CAT- I / CAT- II / Mid-Term

Register No: 22BCE0056 Course Code: BCSE3052 Class NBR: 125-1455
Student's Name: Mr/Ms. Sariphak Ananthi Course Title: Embedded Systems
Programme: B.Tech School: SCOPE Faculty Name: Prof. : Ajitha D
Slot: A1 + TA1 Session: FN/AN/EVN Date of Exam: 27/1/25 No. of pages written:

Ananthi
Signature of the Student

Signature of the Invigilator: Mang
Name / Emp. ID of the Invigilator: 2211
10445

Q No.	Marks	Q No.	Marks	Q No.	Marks	Q No.	Marks	Q No.	Marks
	a b		a b		a b		a b		a b
1		9		17		25		33	
2	2	10		18		26		34	
3	2	11		19		27		35	
4	3	12		20		28		36	
5	4	13		21		29		37	
6		14		22		30		38	
7		15		23		31		39	
8		16		24		32		40	
Sub-Total		Sub-Total		Sub-Total		Sub-Total		Sub-Total	

Grand Total Marks :

11

Marks In Words : one one

Evaluator's Signature RV

Date : 30/1/25

Evaluator's Name : Prof. 18802

Q-3

Timers

Timers are very import for monitor the program throughout the execution,

Watch dog timer when the program is running on the system it is monitored by the ~~watch~~ watch dog timer. If there is any malfunctioning is here in the program it will immediately reset it and stop the execution of the program.

It is very useful to us. as when watch dog has low maintenance and higher efficiency.

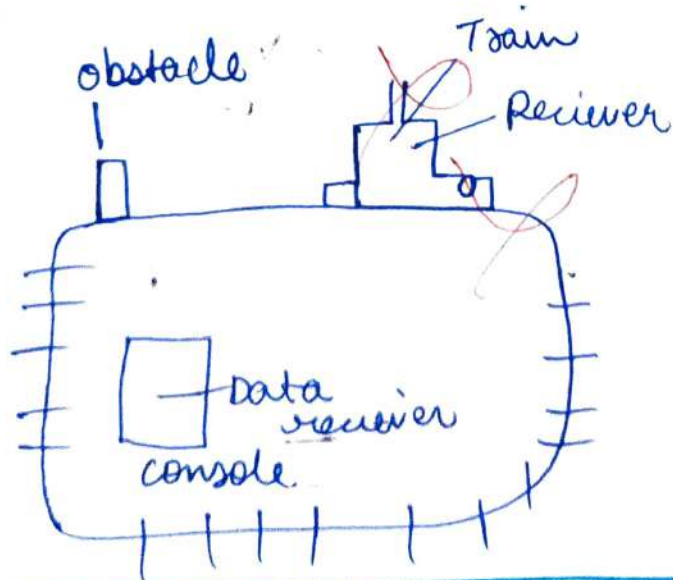
Counters

✓

Q-5

The working principle of proximity sensor is based on Analog to digital circuit where receiver gives the information in the form Analog and converted to digital System for monitoring. The sensor are very useful that they can detect an obstacle ~~to~~ by the sound wave ~~echo~~ echo and by that many hazards ~~that~~ that took place very often can be stopped.

It can be used for developing autonomous vehicle navigation in the vehicle sensor can be used in many ways such as ~~park~~ parking sensor ADAS and Automatic Braking System when any obstacle come in front of the vehicle. GPS ~~is~~ also plays very import role in vehicle navigation, with the help of AI generation we can minimize.



Q-4

ADC

Weighted ADC

This is 8-bit resolution

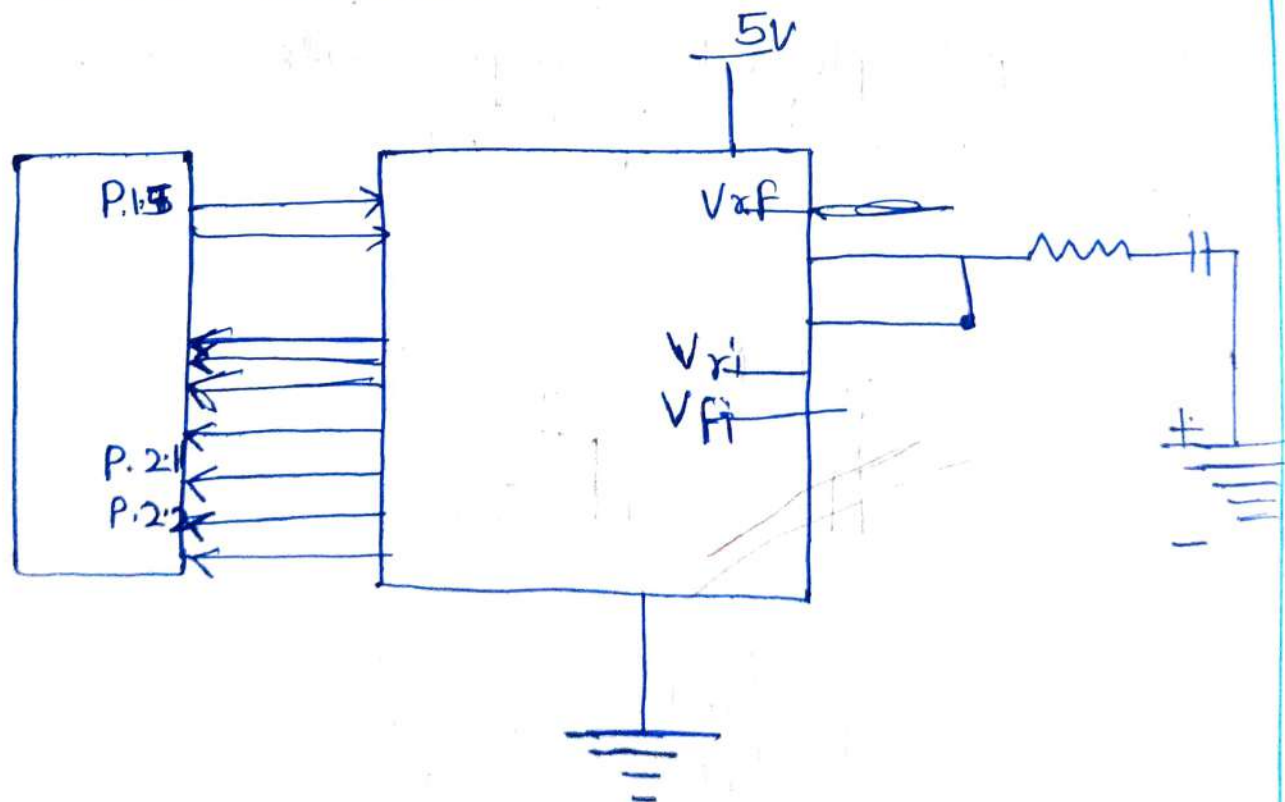
R-2R

In this R-2R multiple registers are connected in parallel.

ADCs are generally used in the ECG ECGs where we get reading in analog we convert into the Digital System for monitoring.

If we take example for sensor work on the ADC only.

ADC are very commonly used like Radio communication, vehicle Automation.



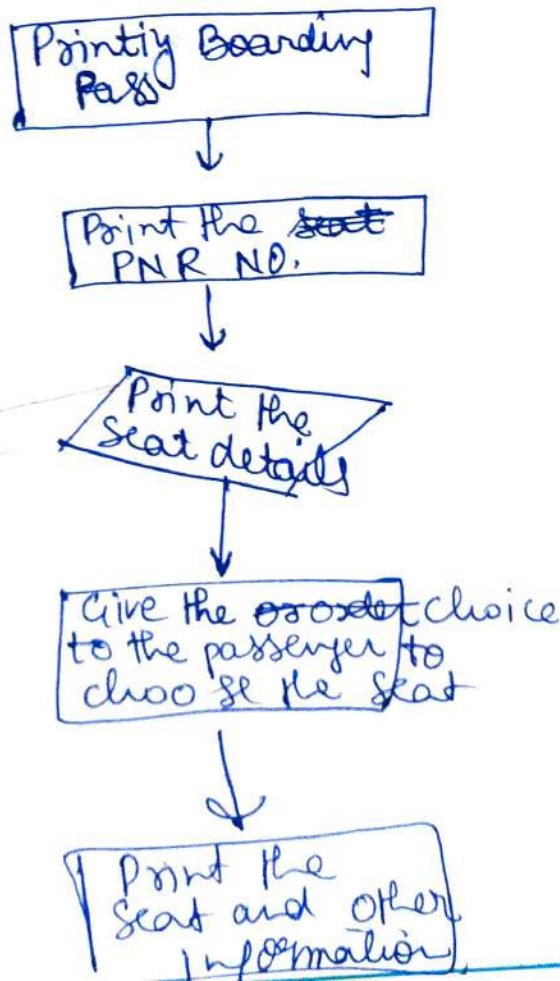
2

Q-2

(i) Logistics

Printing Boarding Pass

Input:- Print the Boarding Pass for the flight



Seat No.

VIT[®]Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)Authorization
Seal

Test : CAT- I / CAT- II / Mid-Term

Register No.: 22BCIO198 Course Code: BCSE305L Class NBR: 1455
 Student's Name: Mr/Ms: Agneepriyadeep Verma Course Title: Embedded Systems
 Programme: B.Tech School: SCOPE Faculty Name: Prof.: Ajitha D
 Slot: A1+TAL Session: FN/AN/EVN Date of Exam: 27.01.25 No. of pages written: 6

Agneepriyadeep Verma
 Signature of the Student

Signature of the Invigilator:

Name / Emp. ID of the Invigilator: 16373

Q No.	Marks		Q No.	Marks		Q No.	Marks		Q No.	Marks		Q No.	Marks	
	a	b		a	b		a	b		a	b		a	b
1	7		9			17			25			33		
2	5		10			18			26			34		
3	7		11			19			27			35		
4	2		12			20			28			36		
5	4		13			21			29			37		
6			14			22			30			38		
7			15			23			31			39		
8			16			24			32			40		
Sub-Total			Sub-Total			Sub-Total			Sub-Total			Sub-Total		

Grand Total Marks :

25

Evaluator's Signature

Date :

21/1/25

Evaluator's Name : Prof.

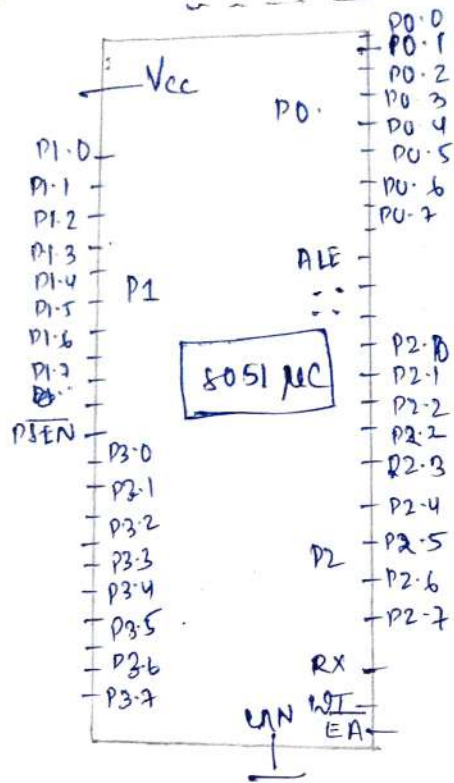
Ajitha D. Jeyaraj

Marks In Words

Two Five

Answers

8-bit Generic Microcontroller (8051)



→ From the physical structure, you can see
↳ it contains various pins including

↳ i) ports - P0, P1, P2, P3

ii) Specific Registers like

- Read RX Pin

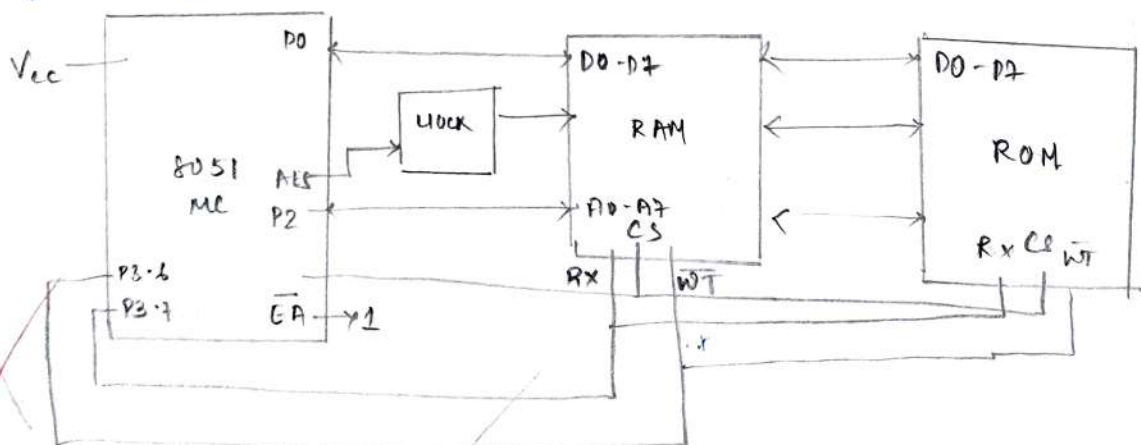
WT Write Pin

EA - External Enable pin

iii) GND - Ground

iv) Vcc - supply voltage (generally 5V).

Since without any external support, 8051 has 4K bytes of internal ROM, 128 bytes of internal RAM. But with the help of 'External Enable' pin, we can extend the capacity of our generic 8051 microcontroller by introducing external ROM (64K bytes) and external RAM (64K bytes) to meet the ~~fast~~ time & space complexity & upgradability.



How External ROM & External RAM Meet the criteria?

i) Upgradability - We ~~can~~ ^{are} upgrade the system to improve its performance, storage, & compute capacity.

ii) Time & Space Complexity - This way, save more space & save our time to make the embedded system an 'real-time' one.

2. i) Logistics

i) Large Computing Power - Since we have to calculate, compute on large dataset, hence need more computing power.

ii) fast calculation & real time delivery - To reduce the time, delay to calculate/compute futuristic in million critical scenarios.

iii) Large storage - To work on very large dataset, its necessary to have large storage too.

iv- space complexity & Time Complexity - working out together, to make the system small in size & Real time Operating manner.

v) - selecting Architecture $\rightarrow$ selecting the best compute model. Some compute Model include -

v) - selecting Programmable language

to which programmable language is suited for complex instruction, & fast execution depends on need. ex - java, Python, C, C#,

etc.

Model include -
↳ state Machine model
↳ object oriented Model
↳ concurrent process Model
etc.

ii) Home Automation

i) less computing Power, Power saving feature - Its need to be ensure, the embedded system does not demand high power to operate in the Home automation.

ii) Large storage - To work on various Home automation system feature, its necessary to implement modularized large code. to implement in harsh environments / unexpected environment.

iii) selecting the Architecture - using von-numann architecture or hardware architecture depends fully on the budget, time & space complexity.

iv) selecting the compute Model - using ^{finite} state Machine, concurrent process, Model, object - Oriented Model.

v) selecting Programming language - selecting the particular language to implement the above - for ex - java, Python, etc.

8. Time delay = 1 sec per cycle

$$(65536 - x) = \frac{1 \text{ sec}}{1.085 \mu\text{s}} = 921658 \text{ sec} \approx 92200 \text{ sec}$$

$922 \times 100 \text{ sec}$

Square wave - 50% duty cycle.

$$x = 65536 - 922 = \text{FC66 H.}$$

To generate the square wave, I will complete P1.0 bit to demonstrate these using timer 0 in mode 1 mode.

```
#include <reg51.h>
```

```
void TONelay();
```

```
void main() {  
    myhit  
    sbit P1_0;
```

```
myhit 220 = 0;    -> to make as input;
```

```
TMOD = 0x02;
```

```
for (i = 0; i < 100; i++) { loop 100 times to make 921 MHz cycle.
```

```
    TONelay();
```

```
    myhit myhit = ~myhit; complement the P1.0 bit
```

```
}
```

```
}
```

```
void TONelay() {
```

```
    TMO = 0x66;
```

```
    TH0 = 0xFC;
```

```
    TR0 = 1;
```

```
    while (TF0 == 0);
```

```
    TF0 = 0;
```

```
    TR0 = 0;
```

```
    return;
```

```
}
```

```
} ; initialize the start value of the timer
```

```
} start the timer
```

```
} stop when timer interrupt flag = 1
```

```
} clear TF0
```

```
} stop timer
```

```
} return from ISR
```

Importance of timer & counters in Embedded system

1) Timer

- It helps to count the time value to work in various applications.
- It is used by watchdog timer to interrupt whether the system is operating or expected or not.
- Generate frequency waves ~~to energy~~ in ~~realtime~~ real time operations.
- used in various applications where the time is critical.
- used in Microwave oven to determine how much time required to heat the food.
- used in Washing Machine to determine for how much minute water will be run inside the washing machine.

2) Counters

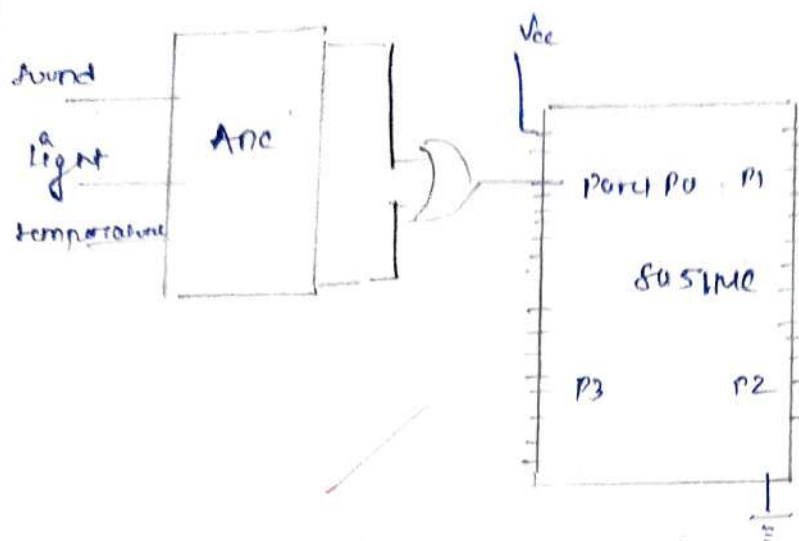
- It helps to count the no. of clock pulses values where required in various applications. Like:-
- Count down timer
- scientific time hour calculations to calculate the no. of clock pulses.

4. Importance of signal conversion in handling mixed signal data

In scientific computers, ALU chip in microcontroller can process and provide output in digital format.

- As time there are different types of signal (Analog signals) present, too complex to handle such a ~~large~~ huge amount of Analog signal will be ~~complex~~ process.

Therefore due to all the above reasons, we need Analog to Digital converter (ADC) to work with Microprocessors with ease.



5. Working Principle of Proximity sensor in Autonomous vehicle Navigation

Proximity sensor takes various ~~log~~ Analog inputs to calculate various things required for autonomous vehicle navigation:-

Functional Requirements.

Light sensor → to detect nearby objects & vehicles, traffic lights etc.

↳ To calculate the speed of the object to adjust the vehicle speed so to avoid collision.

GPS sensor → To track the road, to drive in the correct direction.

Taking the inputs from various sensors it calculates various things -> like -> Vehicle speed, ~~engine~~ nearby vehicle speed

→ collision rate → GPS tracking

→ Virtual environment simulation of the Road / city.

With the help of various actuators like, handle, steering, brake, gear, it controls speed, direction, other features like light, horn, ~~per~~ window panels to ~~sensor~~ give human a safe journey.

24
Seat No.



VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Authorization
Seal

Test : CAT-I / CAT- II / Mid-Term

Register No.: 22BLE2572 ✓ Course Code: BCS305L Class NBR: 1U55
Student's Name: Mr/Ms. P. ARUN Course Title: Embedded System
Programme: Btech School: SCOPE Faculty Name: Prof. : Ajitha D
Slot: AI Session: FN/AN/EVN Date of Exam: 27/1/25 No. of pages written:

Arun. P

Signature of the Invigilator :

Maryn
25/1

Name / Emp. ID of the Invigilator :

10449

Signature of the Student

Q No.	Marks		Q No.	Marks		Q No.	Marks		Q No.	Marks		Q No.	Marks	
	a	b		a	b		a	b		a	b		a	b
1	10		9			17			25			33		
2	8		10			18			26			34		
3	8		11			19			27			35		
4	7		12			20			28			36		
5	10		13			21			29			37		
6			14			22			30			38		
7			15			23			31			39		
8			16			24			32			40		
Sub-Total			Sub-Total			Sub-Total			Sub-Total			Sub-Total		

Grand Total Marks :

43

Marks In Words

four Three

Evaluator's Signature

12

Date :

31/1/25

Evaluator's Name : Prof.

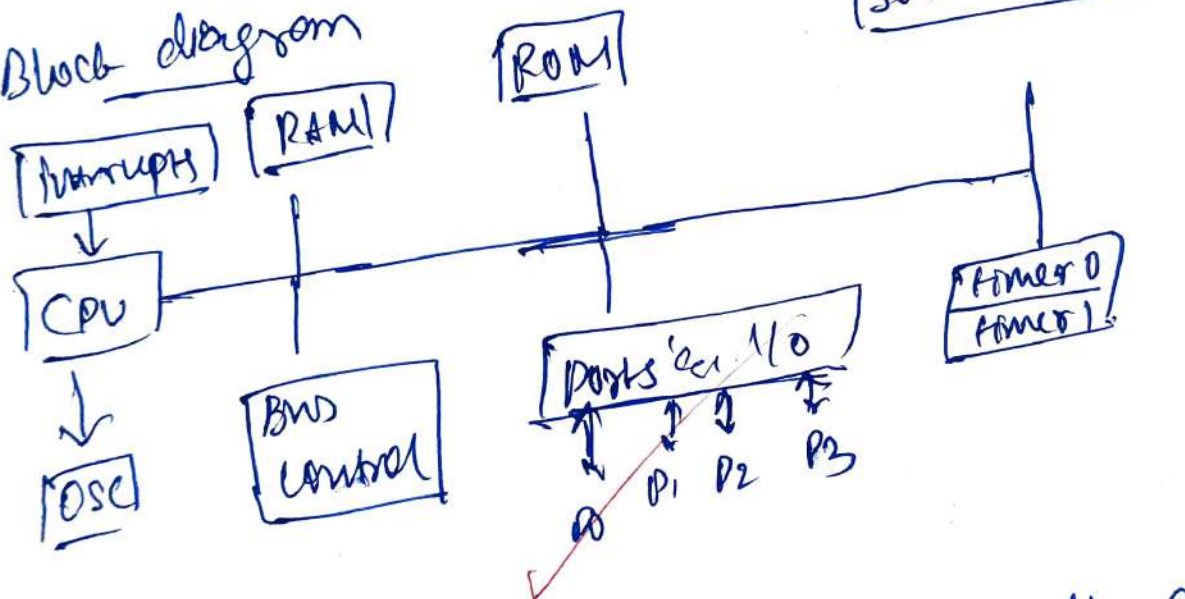
Dr. D. J. S. S.

(Q1) considering the generic 8-bit micro controller - 8051

properties of 8051 MC -

- ⊗ 4KB of ROM
- ⊗ 128B of RAM
- ⊗ 4 serial I/O ports (8 bit each)
- ⊗ 2 timers (16 bit) - timer 0 & timer 1.
- ⊗ 6 interrupts (43 internal, 2 external & 1 reset).

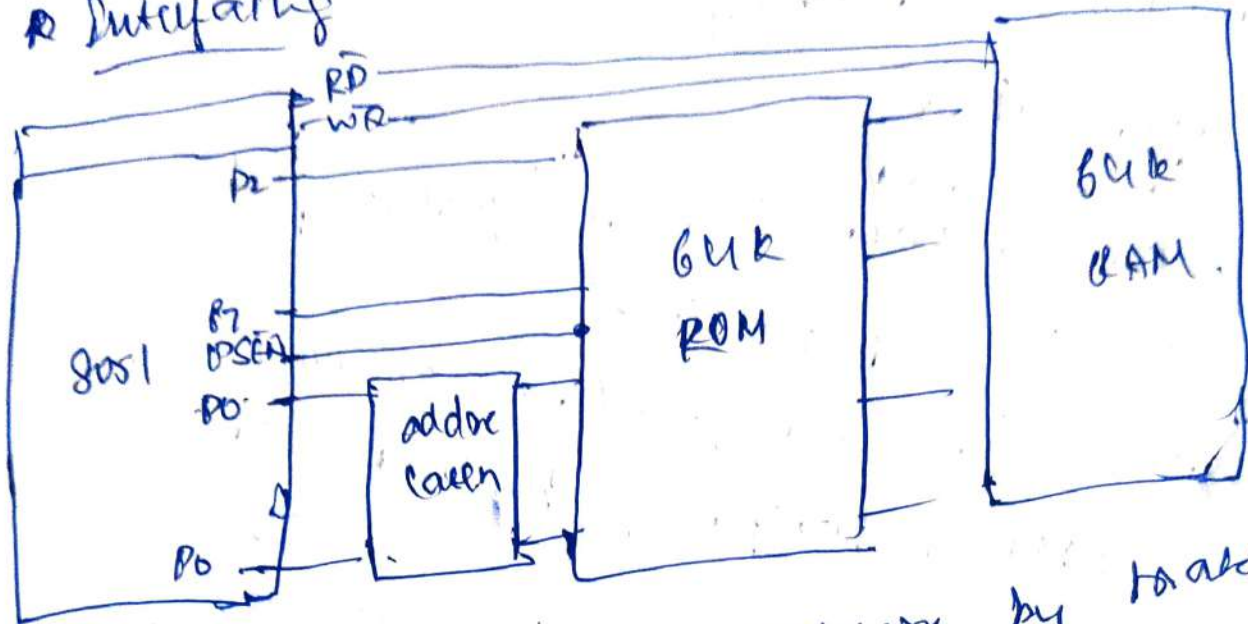
Block diagram



modification:-

- ⊗ Time and space - To increase the available space - we can use external memory interfacing. A total of 64k ROM and 64k RAM can be attached.

Interface



we can interface external memory by adding
 $\overline{EA} = 0$, if we make that then we can
 use full 64K of external memory.
 else if $\overline{EA} = 1$ → 60K of external ROM + 4K
 internal ROM address stand from
 0000H.

Time → using a crystal with higher crystal
 frequency → less seconds the time required
 for 1 clock cycle slowly speed up

Instruction location

(b) upgradability interface

- ⑤ Upgradability -
 - we can ~~connect~~ ^{interface} I/O to port a port to upgrade the performance
 - using ports for parallel communication which enhances transfer of bulk data & program.
 - employ RISC instruction set to upgrade the performance of 8051.
 - using external memory to improve the memory size.
 - RAM 64 K of RAM.
 - complexity - we can use

- we can upgrade the performance
- using ports for parallel communication which enhances transfer of bulk data & program.
- risc instruction set to upgrade the

- employ risc instruction set to upgrade the performance of 8051.
- and preparing to improve the

- employ external memory
- using external memory to improve the performance of CPU.
- the can attain maximum benefit

— using external memory size
memory 64, the can obtain maximum 64K of
RAM. ✓
— the can use

⑤ To improve space complexity - we can use more registers to improve the speed of execution.

(Q2) challenges and issues related to embedded design for (i) Logistics.

Challenges :-

- ① Scalability - MCs are meant to work independently. The processing power and speed are limited / constrained to an embedded system.
- ② In the case of a sudden step spike in logistics, current embedded system would not be able to handle the load with any modification.

- ③ Reliability - Since logistic involve shipping and movement of commodities - the MC has to be reliable for smooth operation, else it may lead to a ~~slow~~ bottleneck and stagnation.

- ④ Lack of resources - for enterprise level of application, we need highly sophisticated MC which are scarce in market.

- ⑤ Security - The MC must have to be secure enough to not leak any sensitive information relating the order company or any confidential info.

⑩ Integration with other system - Logistics involve a very complex intercommunication and design. The MC used ~~in~~ in various ~~of~~ subfields of logistics like supply chain management (for controlling the supply-demand), transportation (automated trucks) need to communicate, but due to their complexity there is no easy integration.

⑪ Logistics Safety - Logistic involving goods movement through automated transport needs to have safety as topmost priority. So the MC present in those automated vehicles should be highly reliable + safe. eg. of MC in automated vehicle - ABS, ADAS ~~etc~~

⑫ Flexibility - The MC should be flexible to address varying workloads, different work type condition eg. on the case of automated truck - it should be flexible to diff road types.

(ii) Home automation -

possible challenges - ① safety, ② security, ③ flexibility,
④ cost, ⑤ suitability.

Explanation -

① safety - MC should have safety as their utmost feature ensuring safety and preventing damages
eg. MC on microwave should have a auto cutoff to turn microwave automatically off when certain time is reached. preventing damages.

② Security - MC installed in device should ensure that the privacy is safeguarded and no compromise should be made on attacks on controller.
eg. Alexa - MC on that device should prevent any unauthorised access.

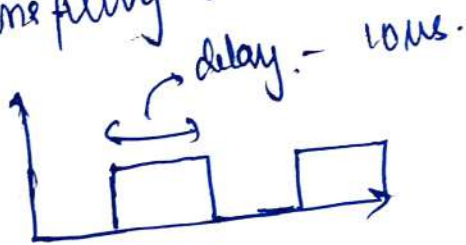
③ flexibility - devices containing MC should ensure that they are flexible to varying workloads and without drop in performance
eg. MC on washing machine should be flexible for various amount of laundry.

- ③ Cost - Intergration of multiple feature would ultimately drive up the cost. Which would limit the number of people who can own it.
- ④ Reliable - Device containing the MC should be built to be reliable. Washing machine present in household ~~it~~ should last for years and not break down early. So Reliability of MC is key.

8

(Q3) Timers - Used to generate delay in a embedded system. It responds to the internal clock pulse of the embedded system. It counts the internal clock pulses and respond accordingly.

eg - To generate a square pulse of width 10 μ s delay. - we use timer to generate the delay & we initialise the timer's value appropriately and when the timer runs down / overflows. - we have successfully created the delay.



Counter - It is used to respond to external stimulus.

It is used to count the external stimulus and then properly display it over a port.

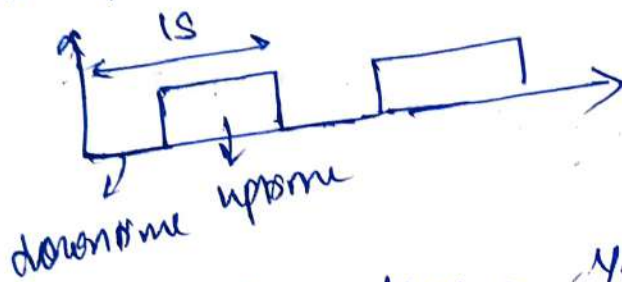
- not dependent on internal clock pulse

eg - when a microcontroller is exposed to a square wave of unknown frequency $\rightarrow$ if we

Apply the external signal to either $\overline{INT0}$ or $\overline{INT1}$ pins (pin P3.2 & P3.3) of 8051, we can use of a counter to count the number of clock pulse thereby obtaining the frequency.

ACP/C

Time period - 1s



$$\begin{aligned} \text{up time} = \text{down time} &= \frac{1}{2} \text{ time period} \\ &= \frac{1}{2} \times 1 = 0.5 \text{ sec} \end{aligned}$$

Control signal freq = 12 MHz

Time for 1 clock pulse $\rightarrow$

$$\frac{1}{12 \text{ MHz}} = 1 \mu\text{s}$$

⑧ number of steps required to count 0.5 s.

$$= \frac{0.5}{1 \times 10^{-6}} \rightarrow \frac{5 \times 10^5}{1} = 500000$$

50,000 $\times 10$

Using timer 0 mode 1 (16-bit).

Representing 60000 in hex $\rightarrow$

Timer start value $\rightarrow$ $65536 - 60000$
 $\rightarrow$ 15536
 $\rightarrow$ 3CB0

Timer value $\rightarrow$ 3CB0, TH0 = 3C, TL0 = B0.

C-code :-

#include <reg51.h>

set mybit = P2^1 // carry square wave in P2^1

void main (void) {

~~unsigned char~~ ~~mybit~~;

~~for~~

while (1) {

mybit = 0

1 // setting bit = 0

delay (1); // calling delay

mybit = 1;

1 // setting bit = 1

1 // calling delay.

delay (1);

}

} void delay (1) {

unsigned char start;
for (start = 0; start < 10; start++) {

TMOD = 0x01; // timer 0 mode 1.

TH0 = 0x3C; //

TL0 = 0xB0;

TR0 = 1; // starting timer 0

while (TF0 == 0); // looping till overflow

TR0 = 0; // reset TR0

TF0 = 0; // reset TF0.

return;

3

8

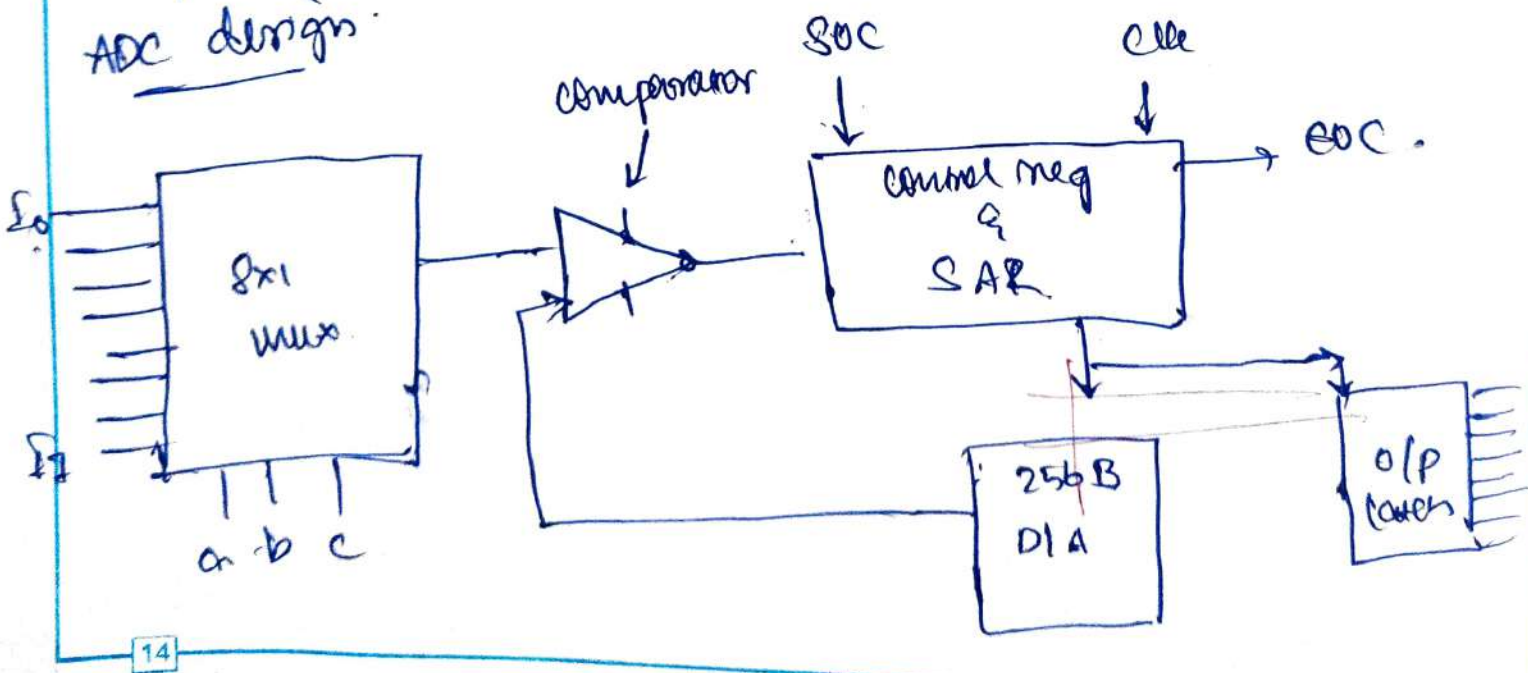
(Ques). we need signal conversion because most of devices use the which can process only digital signal. But the signal in our surrounding are analog in nature.

Similarly there is a need to convert digital to analog signal.

eg - Pressure sensor have to convert the pressure signal (analog in nature) to digital so that necessary computation can be performed.

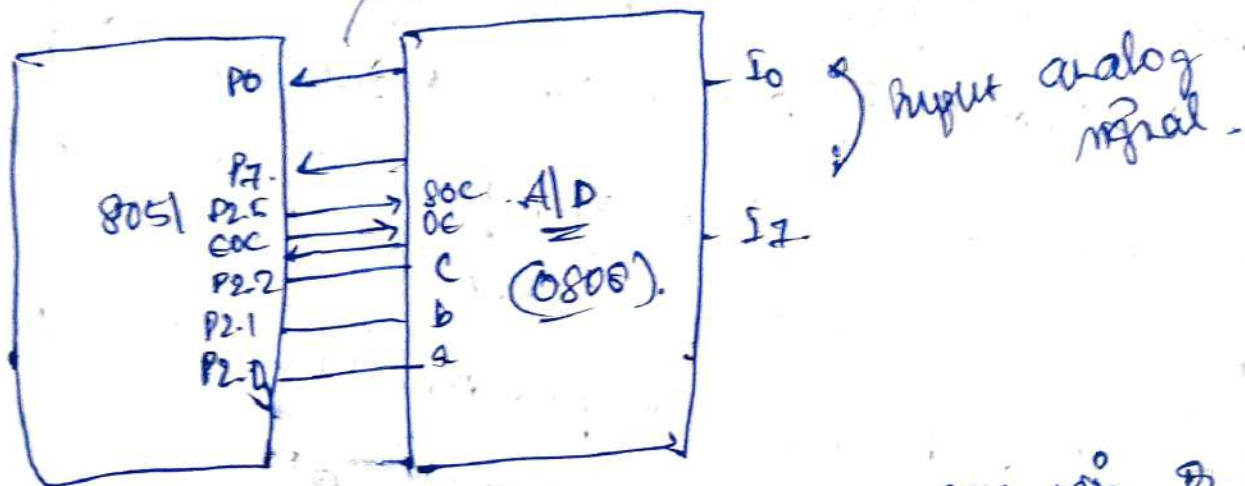
- Music listeners - The songs inside a music player is ~~analog~~ digital in nature, but when it comes out it has to be analog as human can only comprehend analog signals.

ADC design



- The design with Successive Approximation Register (SAR).
- One among the 8 analog signals, only one will be chosen by the select control bits (a, b, c). upon the chosen analog signal, it will be passed through the comparator. The
 - The output from comparator will be given to the control register along with SAR.
 - The SAR generates an digital signal which will be displayed through the latch.
 - It also sends a copy of digital signal to the comparator which again produces the signal and generates a more refined digital signal.
 - This process continues until we reach the digital signal most optimum to the analog signal.
- @ SOC - Start of conversion
 EOC - End of conversion. (active high).
 ST -
 CLK -

Interfacing with MC - digital signals.



- 8051 sends an active high signal to SOC pin of ADC which starts conversion
- SOC sends an active high OE signal to enable MC to send the digital signal.
- Once conversion is over, SOC, which is active high sends $\rightarrow$ low signal signaling completion of.

X

**VIT**

Vellore Institute of Technology

ADDITIONAL SHEET

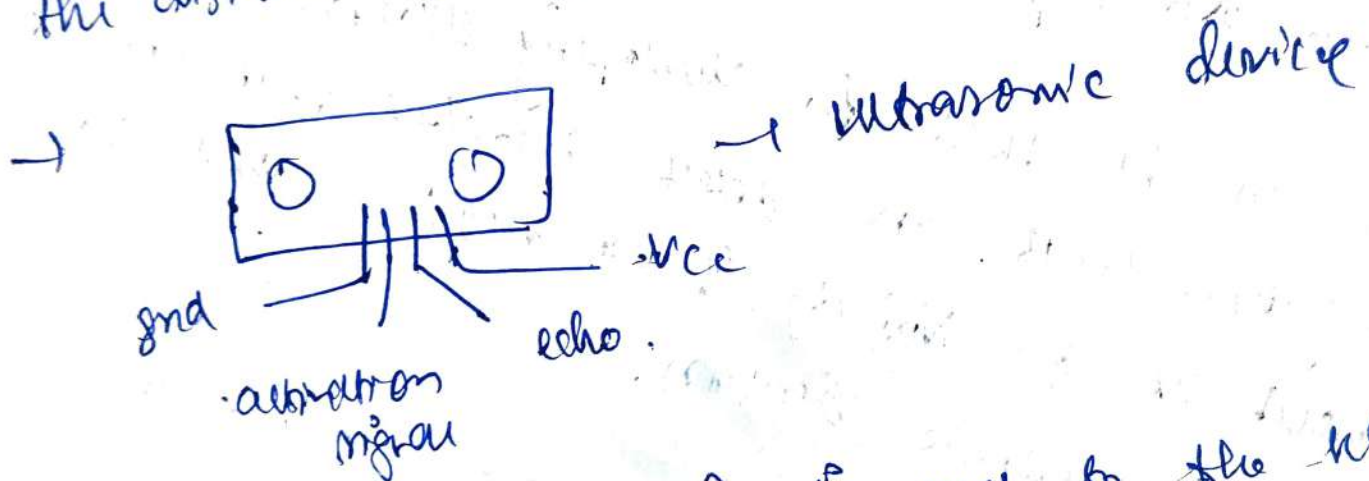
msj
271Initial of the Hall Supervisor with date
Emp. ID.

Q2). Proximity Sensor

- This sensor is used for identifying the objects which are present at the vicinity of the sensor.
- It can be used to determine the distance of the objects from the sensor.
- ② It can be used for developing autonomous vehicle navigation as it can detect vehicles that are present at the vicinity of the vehicle and can perform appropriate decision.
eg:- If the proximity sensor detects a vehicle in front then it can slow down as a response.
- By analyzing the proximity data from the sides of road it can analyze whether to switch lanes.
- It can also assist in automated parking as it detects the car's distance with among other objects and can change the car's path accordingly.

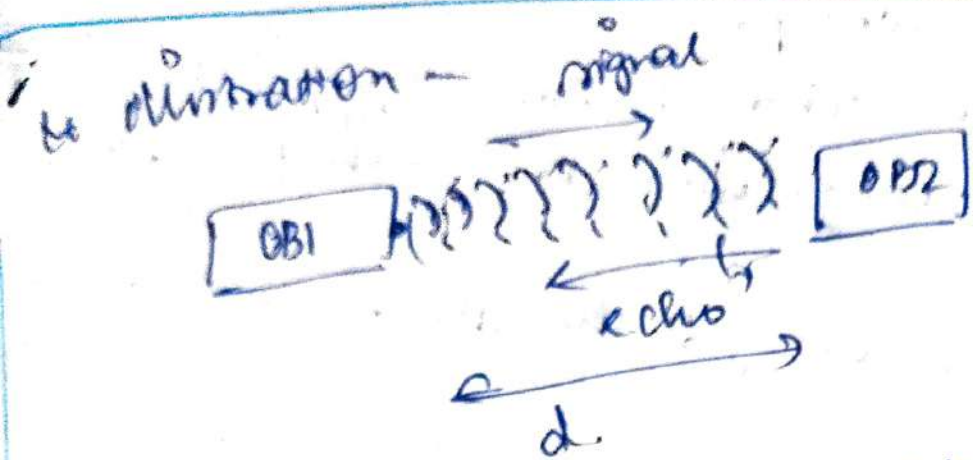
Working in navigation -

The ~~main~~ proximity sensor identifies the distance using ultrasonic waves. It's ultrasonic transducer would send a signal upon getting a activation signal and would wait until its echo reaches back $\rightarrow$ using this approach it can identify the distance of objects.



Once the activation signal is sent to the ultrasonic device $\rightarrow$ it sends a return signal and starts a timer on echo pin, which calculates the time for the signal to reach back.

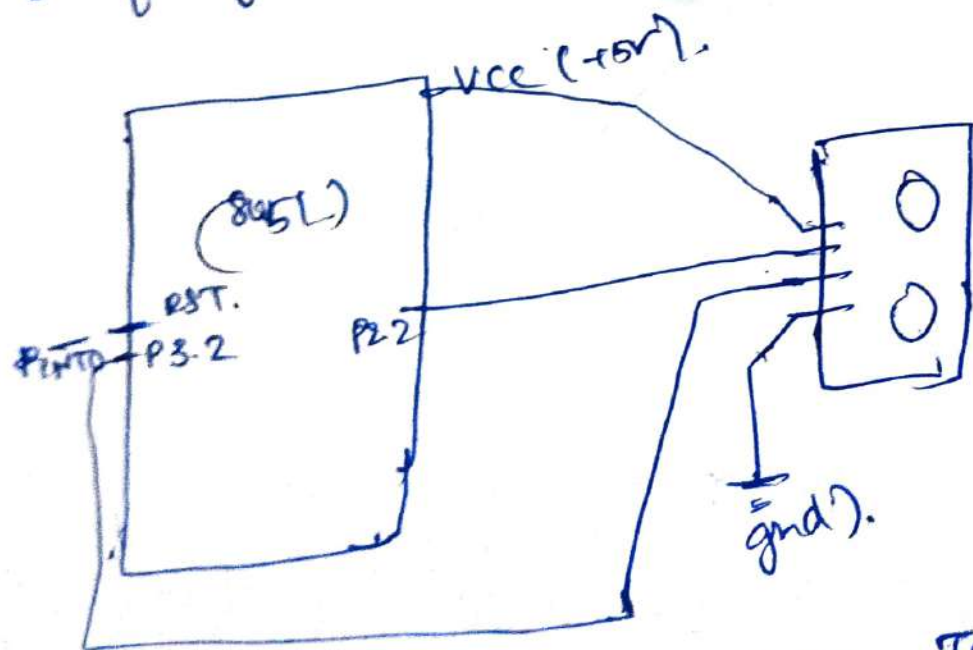
$$\text{dis} = \frac{34300 \times \text{Time}}{2} \quad \rightarrow \quad \frac{34300 \times \text{Timer count} \times 1.045 \times 10^{-6}}{2}$$



$$d = \frac{\text{speed} \times \text{time}}{2} ; \frac{34300 \times 1.085 \times 10^{-1}}{2}$$

from this it calculates the distance and then compares appropriately.

Interfacing with 8051 -



When P2 such the activation, Times with GATE = 1 i.e. so that it can be controlled externally

- is attached to the echo pin.
- as long as the the echo is not returned, the timer sum and timer stops.
 - from the above we get the timer steps / count.

10

VIT
Vellore Institute of Technology

